

Project Title

Blood Donation Network- A MERN Stack Web Application for Blood Donor and Recipient Management

Abstract

Blood Donation Network is a MERN stack web application that connects blood donors, recipients, and hospitals. It allows users to register, search donors by blood group and city, create blood requests, and maintain donation records, making the blood donation process faster and more organized.

Problem Statement

Finding blood donors during emergencies is time-consuming due to the lack of a centralized system. This project provides an online platform to quickly connect donors with recipients and hospitals.

Proposed System

The system provides secure login, donor registration, blood request management, donor search, hospital dashboard, and donation history. It helps users find suitable donors quickly and manage blood requests efficiently.

Objectives

- Provide an online blood donation platform.
- Reduce donor search time.
- Manage blood requests efficiently.
- Maintain donor records securely.

Scope

- Donor Registration
- Blood Request Management

- Hospital Management
- Donation History
- Future support for mobile app and notifications.

Key Features

- User Registration & Login
- Donor Profile Management
- Blood Request Creation
- Search Donors by Blood Group & City
- Hospital Dashboard
- Donation History

Tech Stack

Frontend

- **React.js** – User Interface
- **Vite** – Fast development tool
- **JavaScript** – Programming language
- **TypeScript** - Typed programming language
- **Tailwind CSS** – Responsive styling
- **React Router** – Page navigation

Backend

- **Node.js** – Server-side runtime
- **Express.js** – REST API development
- **JWT** – User authentication
- **bcrypt** – Password encryption

Database

- **MongoDB Atlas** – Cloud database
- **Mongoose** – MongoDB object modeling

Development Tools

- VS Code
- Postman
- Git & GitHub

Deployment

- Frontend – Vercel
- Backend – Render
- Database – MongoDB Atlas

Expected Outcomes

- Faster blood donor search
- Better hospital coordination
- Organized donor records
- Improved blood request management

Conclusion

The Blood Donation Network is a medium-scale MERN web application that simplifies blood donation by connecting donors, recipients, and hospitals. It improves the efficiency of managing blood requests and donor information through a secure and user-friendly platform.